

A CEFR-Inspired Classification Framework with Fuzzy C-Means To Automate Assessment of Programming Skills in Scratch

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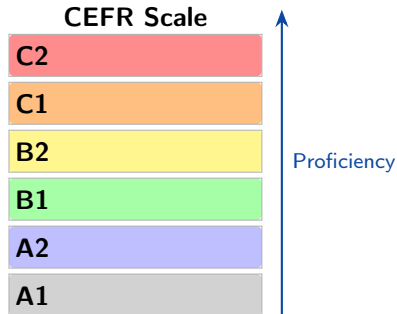
The Problem: Assessing Programming Skills at Scale

? What educators need

- Assess programming proficiency **at scale** (schools, MOOCs, training platforms)
- Not just raw metrics, but **actionable levels**: *where the student is* and *what comes next*
- A framework comparable across institutions

! Gap in the literature

Existing tools (Hairball, Dr.Scratch, LitterBox) produce **discrete metrics** but **no unified, ordered proficiency scale**



Building Blocks of Our Approach



CEFR

- International 6-level scale (A1–C2)
- Already used in **DigCompEdu** for educators
- Precedent: **pycefr** for Python



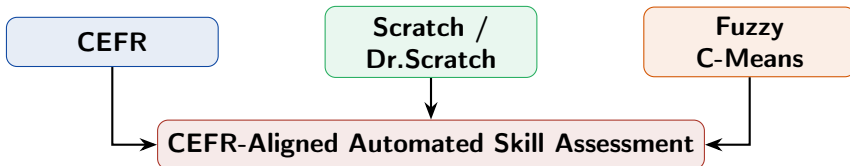
Scratch + Dr.Scratch

- Block-based programming to teach CT
- **9 CT dimensions** scored 0–4
- Dataset: **2,008,246** public projects



Fuzzy C-Means

- Each learner belongs to *multiple* levels simultaneously
- Captures **transitional** states
- Aligns with gradual skill acquisition



Dataset: Over 2 Million Scratch Projects



Dataset overview

- 2,008,246 public Scratch projects
- Automatically evaluated with Dr.Scratch
- Feature vector $\mathbf{x}_i \in \{0, 1, 2, 3, 4\}^9$
- Available on Kaggle (CC BY-SA 2.0)



Train / Test split

- **Train:** 80% = 1,606,596 projects
- **Test:** 20% = 401,650 projects
- Stratified sampling + 5-fold CV

CT Dimension

What it captures

Abstraction	Custom blocks, cloning
Parallelization	Concurrent execution
Logic	Booleans, conditionals
Synchronization	Sprite coordination
Flow Control	Loops, iteration
User Interactivity	Events, user input
Data Representation	Variables, lists
Math Operators	Arithmetic
Motion Operators	Sprite movement

Heavy class imbalance mirrors authentic learning contexts:
Logic / Math concentrate at 0–1; Flow Control more uniform

Why Fuzzy C-Means? Soft Membership for Soft Skills

✗ Hard clustering (K-Means)

Forces each learner into *exactly one* level — discards all information about **in-between states**

✓ Fuzzy C-Means

Assigns **membership degrees** to *all* levels:

$$\mathbf{u}_i = [u_{i,A1}, u_{i,A2}, \dots, u_{i,C2}], \quad \sum_j u_{ij} = 1$$

A student between B1 and B2 is represented *naturally*

⚙️ Parameters (grid search)

- Fuzzification $m = 1.5$ (crisper boundaries)
- Convergence $\varepsilon = 10^{-5}$, max 1000 iter.

Hard clustering

Student → B1 (100%)

Fuzzy C-Means

B1: 55%

B2: 35%

A2: 10%



✓ Captures transitional learning!

Dreyfus model: skill acquisition is *gradual*

Ordinal Mapping: The S_j Criterion

◆ The challenge

FCM produces **unordered** clusters. We need a data-driven way to map them to A1→C2.

💡 Our S_j criterion

For each cluster j , compute the sum of its centroid:

$$S_j = \sum_{d=1}^9 v_{jd}$$

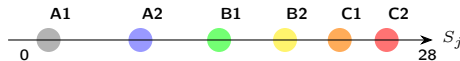
Sort clusters by S_j ascending

⇒ A1, A2, B1, B2, C1, C2

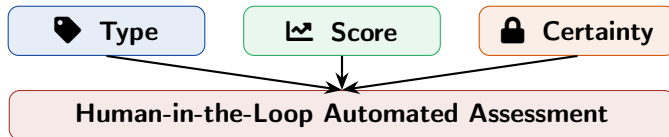
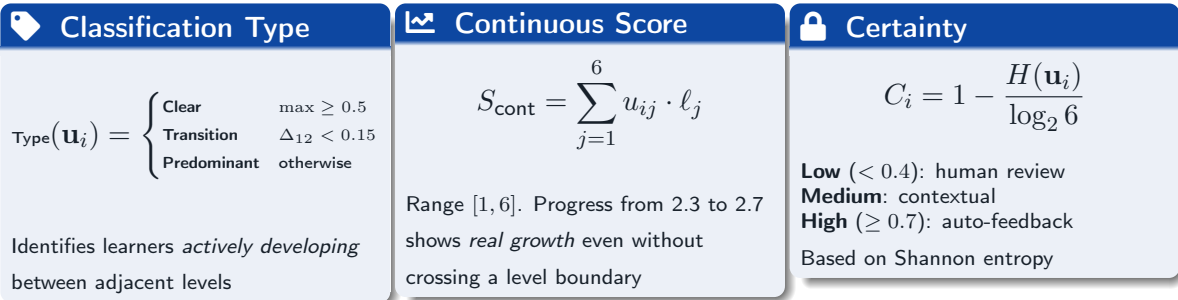
⚖️ Why 6 clusters?

- Domain: aligns with CEFR / DigCompEdu
- Statistical: all pairwise MW $p < 0.001$
- Pedagogical: captures 13.7% transitional learners (lost with $k=2$)

CEFR	S_j	Profile
A1	2.70	Near-zero on all
A2	9.57	Basic across all
B1	15.45	Moderate
B2	20.38	+Logic, Sync, Data
C1	24.47	+Math, Motion
C2	27.98	Near-maximal all



Enhanced Classification: Three Complementary Metrics



Model Performance & Generalization



Near-zero train–test gap

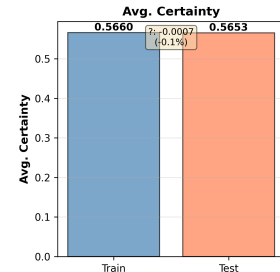
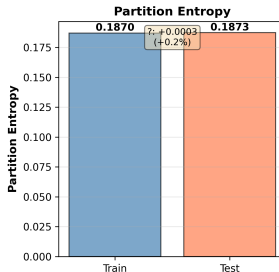
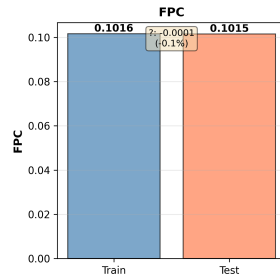
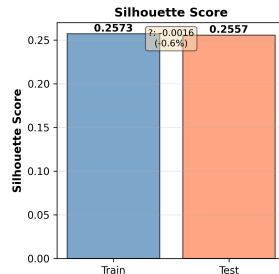
Metric	Train	Test	Δ
Silhouette	0.2573	0.2557	−0.6%
FPC	0.1016	0.1015	−0.1%
Partition Ent.	0.1870	0.1873	+0.2%
Avg. Certainty	0.5660	0.5653	−0.1%

5-fold CV: Silhouette = 0.2631 ± 0.009

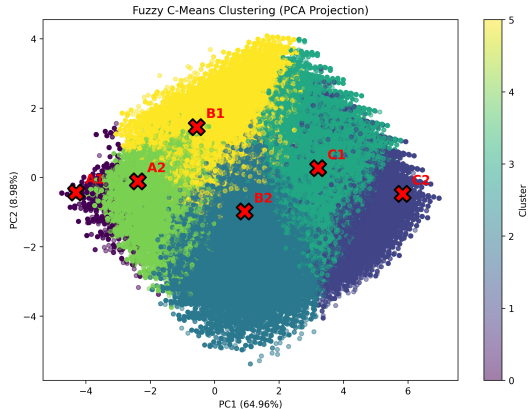


Statistical validation

- Kruskal–Wallis $H = 7,609,964$ ($p < 0.001$)
- All pairwise Mann–Whitney $p < 0.001$
- Global $\tau = 1.000$, $\rho = 1.000$
- Monotonic ordering confirmed in all 9 dims



Cluster Structure: PCA Visualization



PCA reveals the structure

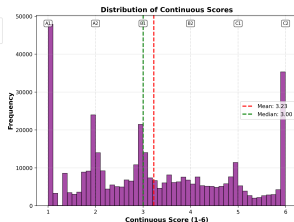
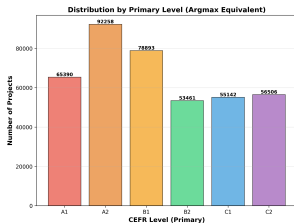
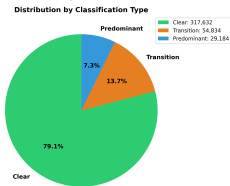
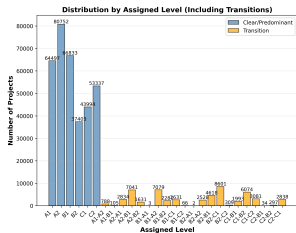
- 2 PCs explain **73.9%** of variance
- Centroids (×) follow a clear ordinal path $A1 \rightarrow C2$
- **Substantial overlap** between adjacent levels
 $\Rightarrow$ confirms the need for fuzzy approach



Strongest ordinal indicators

CT Dimension	τ	ρ
Synchronization	0.966	1.000
Flow Control	0.894	0.986
Logic	0.873	0.971
⋮	⋮	⋮
User Interactivity	0.623	0.771

The B2 Bottleneck: A Curriculum Target



! Key finding

Only **13.3%** reach B2
vs 39.3% at A1–A2 and 27.8% at C1–C2

🧠 Why is B2 hard?

Learners must *simultaneously* integrate:

- Logic (centroid 2.01)
- Synchronization (2.18)
- Data Representation (2.89)

A qualitative cognitive leap vs. earlier levels

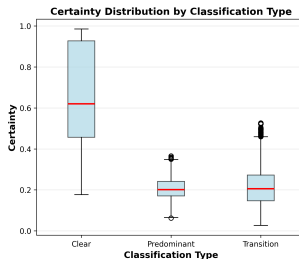
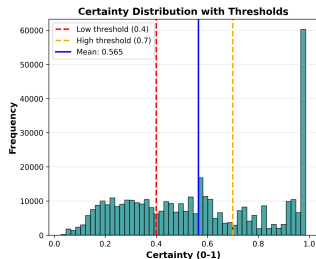
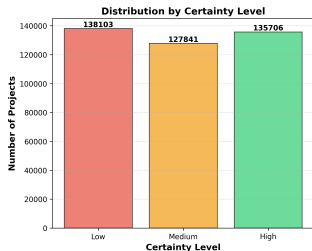
👥 Transition populations

A2–B1: 7,041 learners (1.8%)

B2–C1: 8,601 learners (2.1%)

⇒ Vygotsky's *zone of proximal development*

Certainty as a Human-in-the-Loop Trigger



Balanced distribution

-  **High** (≥ 0.7): 33.8%
auto-feedback
-  **Medium**: 31.8% *contextual*
-  **Low** (< 0.4): 34.4% *human review*

Practical impact

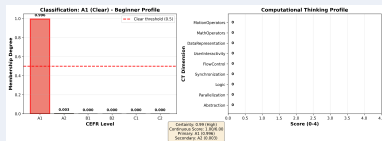
Automating High-certainty cases frees instructors to focus on the **most ambiguous third**
 $\Rightarrow$ *multiplies human impact*

Concrete example

Student “B2–B1 Transition”: strong Motion (4) but weak Synchronization (2). Teacher designs a targeted activity combining both dimensions.

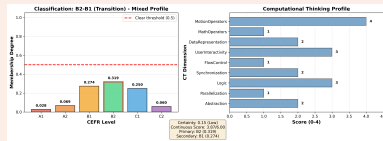
Three Learner Profiles in Practice

A1 — Beginner



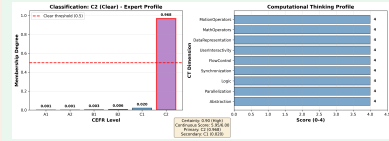
All CT dimensions = 0
Membership: 0.996
Certainty: 0.99 🟢
Score: 1.00 / 6

B2-B1 — Transition



Mixed skill profile
B2: 0.319 | B1: 0.274
Certainty: 0.15 🟡
Score: 3.87 / 6

C2 — Expert



All CT dimensions = 4
Membership: 0.968
Certainty: 0.90 🟢
Score: 5.95 / 6

Table: Algorithm comparison (identical conditions)

Algorithm	Silh.	APN	Usable?
FCM (ours)	0.260	0.097	✓
MiniBatch K-Means	0.293	0.023	!
DBSCAN	-0.156	—	✗

APN (Avg. Proportion of Non-neighbors): higher = better soft-boundary representation

✗ Why not K-Means?

Higher Silhouette but **4× lower APN** — rigid boundaries lose transitional states

✗ Why not DBSCAN?

62 clusters, 11.3% noise, negative Silhouette — unsuitable for ordinal classification

FCM is the *only* algorithm that balances cluster separation **and** soft membership overlap for educational progression

Table: Feedback strategies by level and certainty

Level	Cert.	Suggested Intervention
A1–A2	High	“How could you make this character repeat its dance?” (Flow Control)
B1	High	“What if you needed to remember the player’s score?” (Data)
B2	Any	Trigger synchronization challenge activity
B2–C1	Low	Flag for instructor + peer mentoring
C1–C2	High	“Can you spot repeated patterns to simplify?”



Conceptual dashboard

- **Flagging queue:** Low-certainty students for immediate review
- **Progress timeline:** S_{cont} evolution across submissions
- **Transition alerts:** when students enter ambiguous developmental states



Key design principle

The algorithm **surfaces patterns**.
Pedagogical judgment remains human.

Three main contributions

- 1 **Bridging discrete & continuous:**
fuzzy membership captures **13.7% transitional learners** — invisible in hard clustering
- 2 **Standardized measurement:**
CEFR-aligned via data-driven S_j ordering — comparable across contexts
- 3 **Statistical + pedagogical validity:**
6-cluster solution defensible ($\tau = 1.0$, all MW $p < 0.001$) and educationally meaningful

Key empirical findings

- Silhouette ≈ 0.257 , Certainty ≈ 0.566
- Generalization gap: only **-0.6%**
- **B2 bottleneck** (13.3%)
 $\Rightarrow$ curriculum intervention target
- Strongest indicators:
Sync. ($\tau = 0.966$),
Flow ($\tau = 0.894$),
Logic ($\tau = 0.873$)



Research extensions

- **Longitudinal tracking:**
progression patterns over time
- **Text-based languages:**
Python, JavaScript via AST metrics
- **Multimodal assessment:**
portfolio, debugging, collaboration
- **Cross-cultural validation:**
Scratch's 150+ country community



Institutional integration

- Psychometrically validated
CEFR programming certificates
- Blockchain-backed digital badges
- **DigCompEdu** alignment:
automatic certification module
- Open-source Python/R package
+ public web platform

Immediate next step: adaptive learning system with real-time CEFR classification for dynamic scaffolding & personalised feedback

Thank you!

Questions & Discussion

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Hidalgo-Aragón, R., González-Barahona, J.M., Robles, G. (2026).
*A CEFR-Inspired Classification Framework with Fuzzy C-Means To Automate
Assessment of Programming Skills in Scratch.* CSEDU 2026.